American Computer Science League

2021-2022 ● Contest 2: Short Problem ● Elementary Division Solutions

1. Prefix/Infix/Postfix

To evaluate the prefix expression, group every operator followed by 2 operands or digits at the beginning. Then translate the expression to infix and evaluate from the inside out:

+ / * 5 2 − 7 2 6
= (+ (/ (* 5 2) (− 7 2)) 6)
= (+ (/ 10 5) 6)
= (+ 2 6)
= 8

B. 8

2. Prefix/Infix/Postfix

To evaluate the postfix expression, group every operator preceded by 2 operands or digits at the beginning. Then translate the expression to infix and evaluate from the inside out:

9 3 − 5 2 + * 2 3 ^ 3 2 ^ + −
= (((9 3 −) (5 2 +) *) ((2 3 ^) (3 2 ^) +) −)
= ((6 7 *) (8 9 +) −)
= (42 17 −)
= 25

D. 25

3. Prefix/Infix/Postfix

To convert the following infix expression to prefix, use the order of operations (PEMDAS)and write each infix expression with the operator before both operands in the same order:

$$\frac{4 * (7 - 5)}{(8 - 2) / (2 + 1)}$$

= (4 * (− 7 5)) / ((− 8 2) / (+ 2 1))
= (* 4 − 7 5) / (/ − 8 2 + 2 1)
= / * 4 − 7 5 / − 8 2 + 2 1

E. / * 4 − 7 5 / − 8 2 + 2 1

4. Prefix/Infix/Postfix

To convert the following infix expression to postfix, use the order of operations (PEMDAS)and write each infix expression with the operator after both operands in the same order:

5 * 7 − (1 + 4) * 6 + 2
= ((5 7 *) − ((1 4 +) * 6)) + 2
= ((5 7 *) − ((1 4 +) 6 *)) + 2
= ((5 7 *) ((1 4 +) 6 *) −) + 2
= ((5 7 *) ((1 4 +) 6 *) −) 2 +
= 5 7 * 1 4 + 6 * − 2 +

A. 5 7 * 1 4 + 6 * − 2 +

5. Prefix/Infix/Postfix

To convert the following expression from prefix to postfix, change each group of 2 operands preceded by an operator to a group of 2 operands succeeded by an operator::

− / * ^ 5 2 4 2 + 1 9
= − / * (5 2 ^) 4 2 (1 9 +)
= − / ((5 2 ^) 4 *) 2 (1 9 +)
= − (((5 2 ^) 4 *) 2 /) (1 9 +)
= 5 2 ^ 4 * 2 / 1 9 + −

B. 5 2 ^ 4 * 2 / 1 9 + −